

Proposed Solution

Ideation Phase | Import Data Using Transform Maps

1. Solution Summary

Build a repeatable import process in ServiceNow. External files are loaded through a **Data Source** into an **Import Set** staging table. A **Transform Map** then maps, cleans and writes the data into the target table, using a **coalesce** field to update existing records instead of creating duplicates. Results are verified in **Transform History**.

2. High-Level Flow

Step	Component	What happens
1	Source file	CSV/Excel prepared with a header row
2	Data Source	Defines file type, sheet and header row
3	Import Set table	Raw rows loaded into a staging table (u_ prefix)
4	Transform Map	Source fields mapped to target fields; scripts and lookups applied
5	Coalesce	Existing record matched on unique field and updated, else inserted
6	Target table	Clean data stored in sys_user, incident, cmdb_ci or custom table
7	Transform History	Counts of inserted, updated, ignored and error rows reviewed

3. Key Features

- **Field mapping:** direct, scripted and reference mappings between staging and target fields.
- **Coalesce:** prevents duplicates and allows safe re-loads.
- **Transform scripts:** onStart, onBefore, onAfter and field map scripts for cleaning and validation.
- **Error handling:** invalid rows are ignored and logged, not silently loaded.
- **Reusability:** same map is reused for every future file with the same layout.

4. Example Field Mapping

Source (staging)	Target	Method
u_first_name	first_name	Direct
u_last_name	last_name	Direct
u_email	email	Direct, coalesce
u_department	department	Reference lookup on name
u_status	active	Script (Active = true)

5. Novelty and Value

- Uses standard ServiceNow features with no extra licences.
- Reduces manual effort and removes duplicate data.

- Gives a complete audit trail of every import.
- Scales from a few rows to thousands of records.

6. Feasibility

Area	Assessment
Technical	Fully supported by Import Sets and Transform Maps in a ServiceNow PDI
Operational	Simple for admins; data owners only follow a file template
Cost	No additional licence or tooling cost
Risk	Low, when tested with sample files before real data

7. Success Criteria

- 100% of valid rows loaded to correct fields.
- Zero duplicate records after repeated loads.
- Rejected rows clearly identified in logs.
- Import time reduced significantly compared with manual entry.